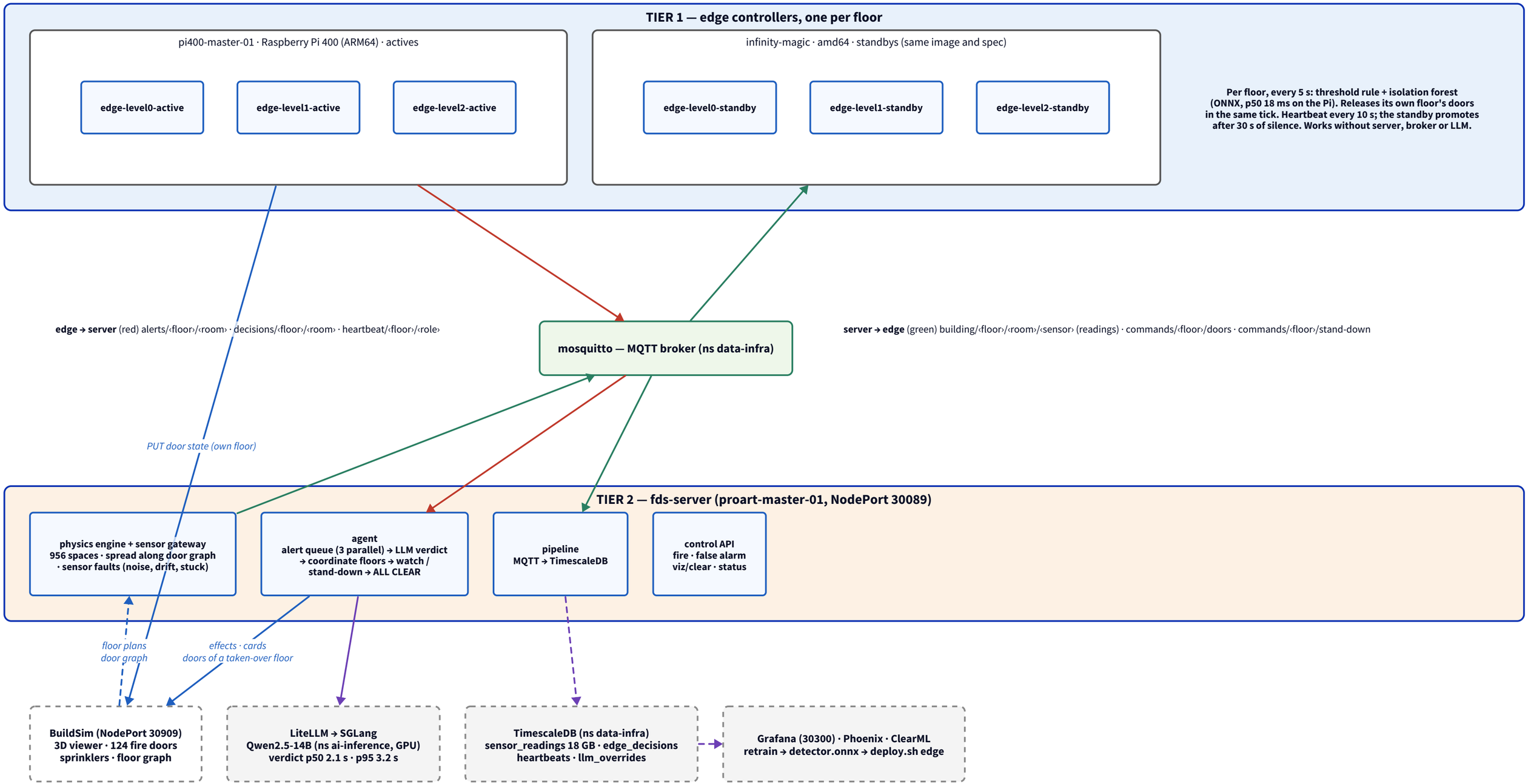

TIER 1 — edge controllers, one per floor
pi400-master-01 · Raspberry Pi 400 (ARM64) · actives
edge-level0-active
edge-level1-active
edge-level2-active
infinity-magic · amd64 · standbys (same image and spec)
edge-level0-standby
edge-level1-standby
edge-level2-standby
Per floor, every 5 s: threshold rule + isolation forest (ONNX, p50 18 ms on the Pi). Releases its own floor's doors in the same tick. Heartbeat every 10 s; the standby promotes after 30 s of silence. Works without server, broker or LLM.
edge → server (red) alerts/‹floor›/‹room› · decisions/‹floor›/‹room› · heartbeat/‹floor›/‹role›
mosquitto — MQTT broker (ns data-infra)
server → edge (green) building/‹floor›/‹room›/‹sensor› (readings) · commands/‹floor›/doors · commands/‹floor›/stand-down
PUT door state (own floor)
TIER 2 — fds-server (proart-master-01, NodePort 30089)
physics engine + sensor gateway
956 spaces · spread along door graph
· sensor faults (noise, drift, stuck)
agent
alert queue (3 parallel) → LLM verdict
→ coordinate floors → watch /
stand-down → ALL CLEAR
pipeline
MQTT → TimescaleDB
control API
fire · false alarm
viz/clear · status
floor plans
door graph
effects · cards
doors of a taken-over floor
BuildSim (NodePort 30909)
3D viewer · 124 fire doors
sprinklers · floor graph
LiteLLM → SGLang
Qwen2.5-14B (ns ai-inference, GPU)
verdict p50 2.1 s · p95 3.2 s
TimescaleDB (ns data-infra)
sensor_readings 18 GB · edge_decisions
heartbeats · llm_overrides
Grafana (30300) · Phoenix · ClearML
retrain → detector.onnx → deploy.sh edge